

industrial, ITE and household designs without requiring additional cooling. The RACM550 series is designed to support up to 300W of continuous output power at 24V without cooling fans. Building on RACM230 230W series, this highly efficient baseplate-cooled design measures just 38mm total height and enables direct heat dissipation through metal housings in the end-application.

RECOM Asia Pte Ltd
www.recom-power.com

EV charger solution

To charge batteries more efficiently and to ensure their long life, a smart battery management or charging system is needed. To realise such an electric vehicle (EV) charging system, Holtek has come up with smart EV charger solutions based on its low cost, ASSP flash MCUs HT45F5Q-X for charging batteries in EVs. At present, three EV charger designs suitable for Indian market with specifications of 48V/4A, 48V/12A and 48V/15A are available, to achieve faster development of the product. This semiconductor-based smart charging system is an ideal solution for charging EV batteries, and can support both lithium-ion and lead acid battery types.

Holtek Semiconductor India Pvt Ltd
www.holtek.com

MISCELLANEOUS

Automotive central control units

Tieto and STMicroelectronics have announced their collaborative efforts to develop central control unit (CCU) software to run on STMicroelectronics'



popular Telemaco3P platform. Accelerating electrification and connectivity requirements of vehicles are pushing for greater and greater processing power and uncompromised cybersecurity. In turn, automakers' demands for central control units meeting connectivity, data privacy, security and over-the-air update needs have led ST to develop Telemaco family of automotive, multi-processor SoCs (systems-on-chips) and its associ-

ated Telemaco3P Modular Telematics Platform (MTP), which provides an open development environment for prototyping advanced smart driving applications. The secure and safe Telemaco3P automotive SoC is the industry's first microprocessor to embed an isolated hardware security module for state-of-the-art security. The module provides safety measures for implementation of ASIL-B-qualified systems.

STMicroelectronics India
www.st.com, www.tieto.com

Camera solution

Analog Devices (ADI) has announced a collaboration with Jungo to develop a time-of-flight (ToF) and 2D infrared (IR)-based camera solution to enable driver- and in-cabin monitoring in vehicles. The combination of ADI's ToF technology with Jungo's Codriver software is expected to enable monitoring of vehicle occupants for levels of drowsiness and distraction by observing head and body position as well as eye gaze. The solution is also expected to enable smart vehicle interaction based on face, body and hand gestures, and offer a facial recognition capability that identifies individuals in the cabin and gives them access to features such as infotainment personalisation, personalised services and ride-sharing payments.

Analog Devices Inc.
www.analog.com

Attendance solution

Matrix COSEC time-attendance solution promotes a smart and productive workplace. It is a one-stop solution for managing and controlling attendance data. The solution is designed to meet the diverse requirements of any organisation by offering a superlative range of flexibility in attendance policies, configuration, integration and report generation. Additionally, it offers the following features:

- Multiple credentials - mobile, biometric, RFID cards
- Real-time monitoring
- Comprehensive reports
- Real-time alerts and notifications
- Seamless integration with payroll, ERP or HRM software

Matrix Comsec
www.matrixcomsec.com,
www.matrixaccesscontrol.com

Atomic clock

As reliance on precise frequency and timing increases due to Global Navigation Satellite System (GNSS) enabling 5G communication networks, data centres and other mission-critical infrastructure, smaller size and high-performance atomic clock technology has become essential to support both military and commercial applications. To meet demand for a small footprint atomic clock, Microchip Technology has announced the industry's highest performance atomic clock for its size and power. The new device delivers a wider thermal range, critical performance improvements and other enhancements over previously available technology. The next-generation MAC-SA5X miniaturised rubidium atomic clock produces a stable time and frequency reference that maintains a high degree of synchronisation to a reference clock, such as a GNSS-derived signal.

Microchip Technology Inc.
www.microchip.com

Single board computer

Powered by NXP i.MX8 processor, iWave single board computer (SBC) is equipped with up to eight Armv8 64-bit



heterogeneous CPU cores (2x Arm Cortex-A72, 4x Arm Cortex-A53 and 2x Arm Cortex-M4F). This highlights the extremely powerful computational capability of the SBC, making it an awaited platform for unleashing complex to implement applications with utmost efficiency and ease. Advance use-cases include Edge computing, industrial automation, 4K imaging, machine vision, the Industrial Internet of Things (IIoT), robotics, augmented and virtual reality (AR and VR), and drones.

iWave Systems
www.iwavesystems.com